NUS · DENTAL MATERIALS · LECTURE 03

Polymers & Polymerization

Student handout — from monomer chemistry to clinical shrinkage.

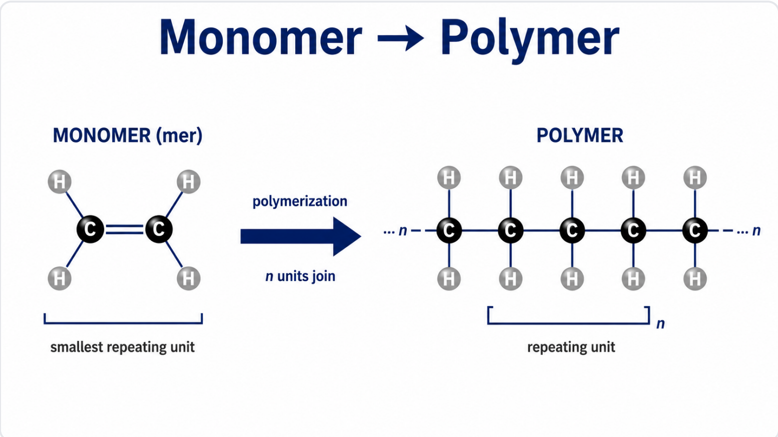
Revision aid, not a full transcript of the lecture. Organised on the structure → property → performance spine.

STRUCTURE

What polymers are made of

Vocabulary

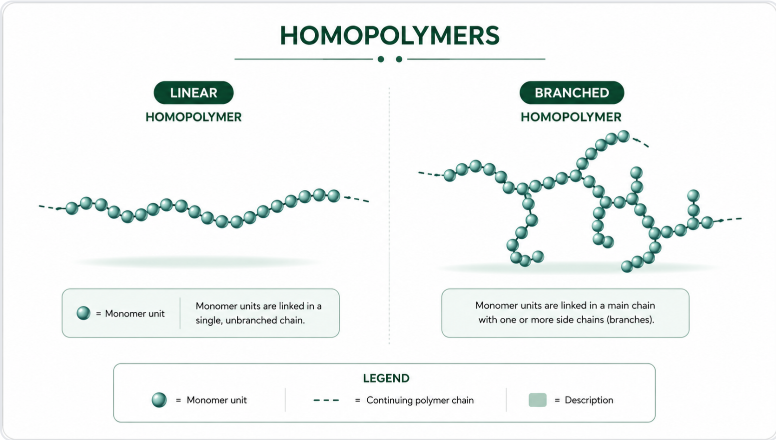
- **Monomer** — a small reactive molecule (one "mer" unit). **Oligomer** — a few units. **Polymer** — many units covalently linked into long chains.
- **Degree of polymerization (n)** = number of mer units in the chain.
- **Molecular weight**: MW = (MW of one mer) × n. Higher MW ⇒ longer chains ⇒ stronger, stiffer, higher softening temperature.
- By composition: **homopolymer** (one monomer), **copolymer** (two), **terpolymer** (three) — mixing monomers tailors properties.



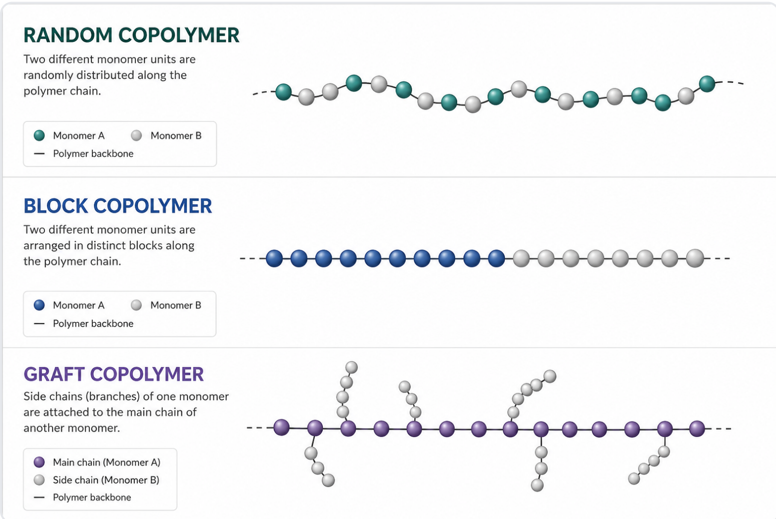
Many monomers link covalently into a long chain — the polymer.

Chain architecture

- **Linear / branched / cross-linked**: how chains are connected sets mobility. Linear & branched chains can slide (they melt/soften = **thermoplastic**); cross-linked chains are locked by covalent bridges into a network (**thermoset** — degrades before it melts).
- Thermoset ≠ "just harder": the difference is a covalent **network**, not stiffness alone.
- Copolymer sequence — **random / block / graft** — changes flexibility, compatibility and clarity.



Linear vs branched architecture — both can soften with heat (thermoplastic).



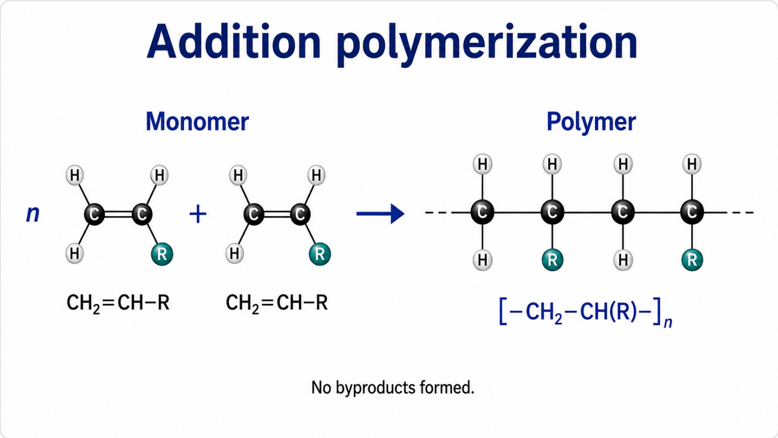
Copolymer arrangements: random, block, graft.

PROPERTY

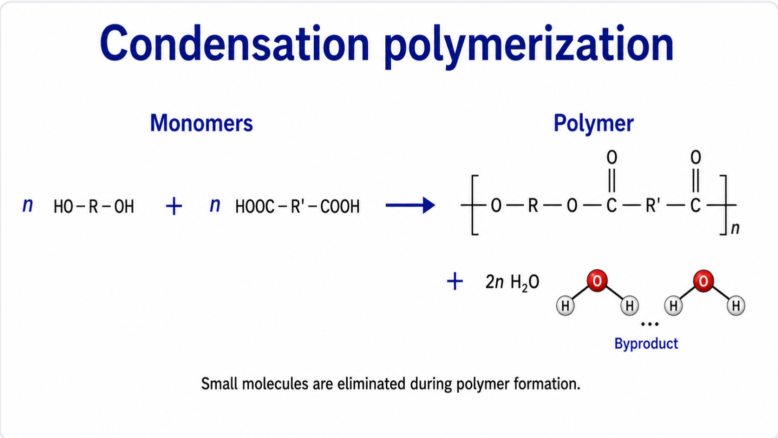
How polymers form & what controls behaviour

Addition vs condensation

- **Discriminator = the by-product. Addition** polymerization: monomers add on with **no by-product** → dimensionally stable. **Condensation**: each bond ejects a small molecule (e.g. water/alcohol) → shrinkage/instability as by-product escapes.
- Clinical anchor: **addition silicone (PVS)** vs **condensation silicone** — same “silicone”, opposite dimensional stability because of the by-product. PVS is far more accurate over time.



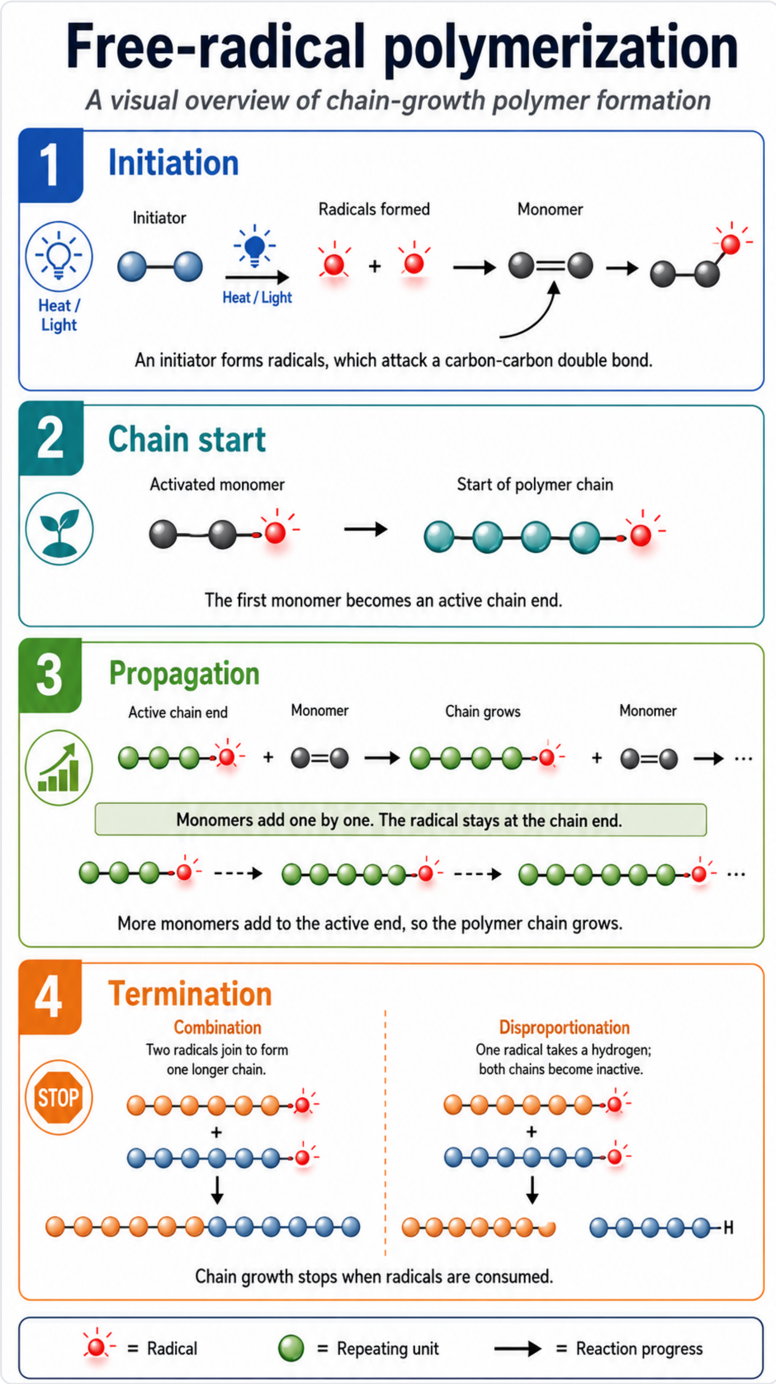
Addition: monomers join with no by-product → dimensionally stable.



Condensation: each linkage releases a small molecule → shrinkage.

Free-radical polymerization (addition-type; dental resins)

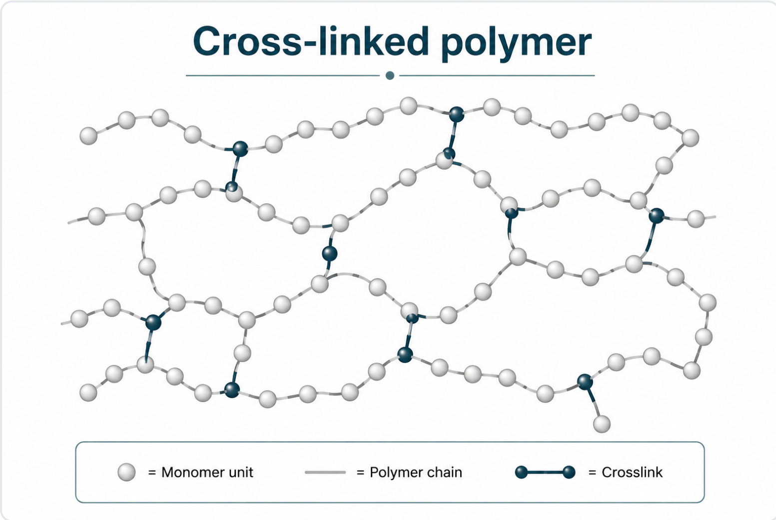
- Three stages: **initiation** (initiator forms radicals — e.g. camphorquinone + light, or benzoyl peroxide) → **propagation** (chain grows, radical stays at the tip) → **termination** (two radicals meet).
- **Oxygen inhibits** the surface → the sticky oxygen-inhibited layer. **Inhibitors** give shelf life; **light** triggers cure on demand.
- Never 100%: leftover unreacted monomer = **residual monomer** (biocompatibility/ leachables). Distinguish **degree of conversion** (fraction of C=C reacted) from **degree of polymerization** (chain length).



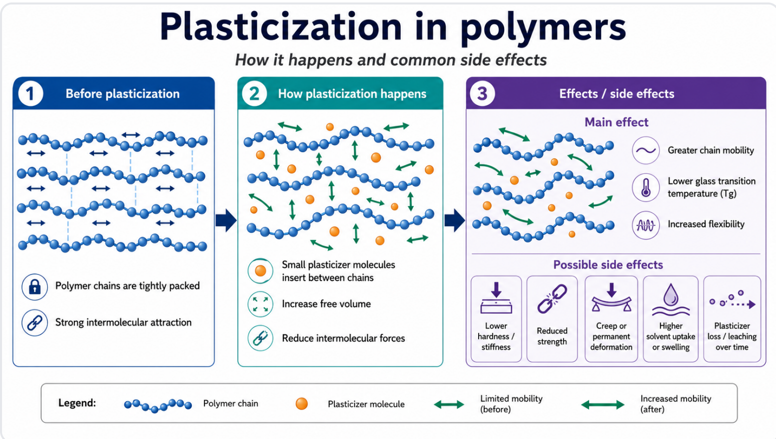
The three stages of free-radical polymerization.

Predicting properties

- ↑ **Molecular weight** → ↑ strength, stiffness, softening temperature.
- ↑ **Cross-linking** → ↑ stiffness, ↑ solvent/water resistance, ↓ swelling — but **too much → brittle**.
- **Plasticizer** → slips between chains, ↑ mobility, ↓ glass-transition temperature (Tg) → softer, more flexible (e.g. soft liners), but can leach out over time.



Cross-links: covalent bridges lock chains into a network.

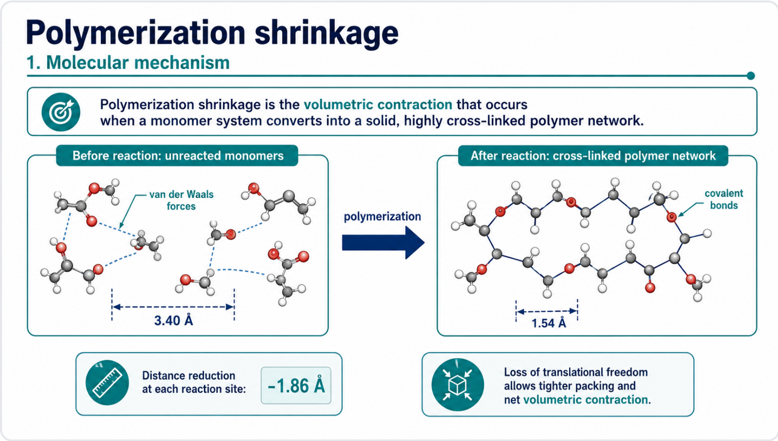


Plasticizer separates chains → more mobility, lower Tg.

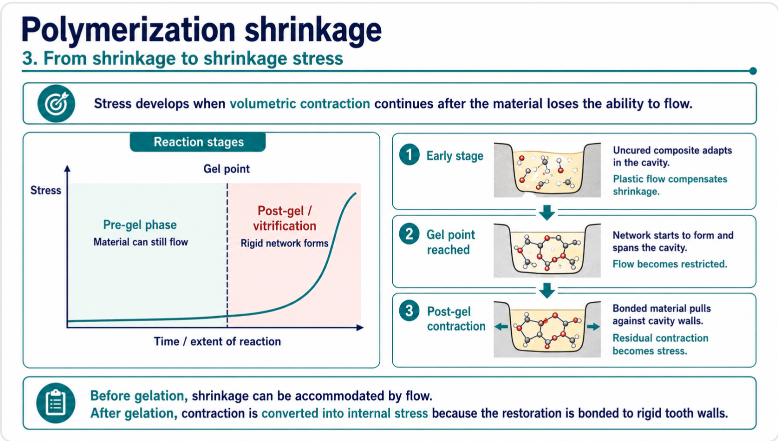
PERFORMANCE

Shrinkage, stress & clinical failure

- **Why it shrinks:** monomers held apart by weak van der Waals forces (~3.4 Å) are pulled into short covalent bonds (~1.5 Å) as chains form → the mass occupies less volume.
- Dental resin composites shrink about **1.5–6 % by volume** (matrix chemistry, e.g. Bis-GMA/TEGDMA, and filler load).
- **Pre-gel vs post-gel:** before the gel point the material can flow and relieve strain; after gelation it is rigid, so further shrinkage builds **stress**.
- **Shrinkage ≠ stress.** The clinical problem is **post-gel stress**: if it exceeds bond strength → debonding, marginal gap, microleakage, post-op sensitivity, secondary caries.
- **C-factor** (bonded ÷ unbonded surfaces): higher C-factor cavities (e.g. deep Class I) trap more stress.
- **Manage it:** incremental layering, adequate cure, and stress-relieving/low-shrink materials.



Van der Waals spacing (~3.4 Å) → covalent bond (~1.5 Å): the molecular cause of shrinkage.



Post-gel: the set material can no longer flow, so shrinkage becomes stress at the margin.

Take-home

- Structure (chain length, architecture, cross-linking) predicts properties; the by-product tells you if a reaction is stable.
- Dental resins cure by free-radical addition — never fully; residual monomer and post-gel stress are the recurring clinical themes.

Dental Materials Science · Lecture 03 handout · structure → property → performance.